

Within-family consistency and the descriptive fit-matrix

Of 13 evaluated family × task cells, 12 agree; the lone within-family split is the T2 donor OT pair.

DESCRIPTIVE / EXPLORATORY

PREREG rule 5, not a hypothesis test

a Within-family verdict agreement

cell value: ρ = within-family per-unit Spearman concordance

CONTEXT-TRANSFER			UNSEEN-PERTURBATION		
T1 cell-context	T2 donor	T5 cell-context	T4 modality	T3 unseen-pert	T5 unseen-cpd
Latent	none beat $\rho=+0.26$	none beat $\rho=+0.37$	none beat $\rho=-1.00$ PENDING $n<3$ units	none beat $\rho=+0.70$	none beat ρ n.a.
Graph			none beat $\rho=+0.90$ PENDING $n<3$ units	none beat $\rho=-0.30$	
Foundation				none beat $\rho=+0.70$	
Hybrid					none beat ρ n.a.
Chemistry					
OT		SPLIT CellOT beats scPRAM fails	none beat ρ n.a. PENDING single-seed		

OT-family donor exception

the lone donor win is
CellOT-specific, not family-level

CellOT 0.367

scPRAM 0.159

paired $n = 106$ shared donors
scPRAM wins 2/106 gap -0.207
Wilcoxon $p = 5.5 \times 10^{-19}$

b Descriptive fit-matrix: does the family beat the universal floor?

cell value: best-member gap vs universal floor (CI_{lo} = cluster-bootstrap lower bound)

CONTEXT-TRANSFER			UNSEEN-PERTURBATION		
T1 cell-context	T2 donor	T5 cell-context	T4 modality	T3 unseen-pert	T5 unseen-cpd
Latent	-0.024 $CI_{lo} -0.048$	-0.068 $CI_{lo} -0.078$	-0.090 $CI_{lo} -0.160$	-0.102 PENDING $n<3$ units	-0.372 $CI_{lo} -0.519$
Graph			-0.153 PENDING $n<3$ units	-0.258 $CI_{lo} -0.309$	
Foundation				-0.284 $CI_{lo} -0.394$	
Hybrid	-0.040 $CI_{lo} -0.051$	-0.209 $CI_{lo} -0.256$	-0.636 PENDING $n<3$ units	-0.479 $CI_{lo} -0.538$	
Chemistry		+0.119 $CI_{lo} +0.101$			-0.008 PENDING $n<3$ units
Shift			-0.044 PENDING $n<3$ units		

beats floor ($CI_{low} > 0$) consistent: none beat within-family split pending ($n<3$ / single-seed) not evaluated